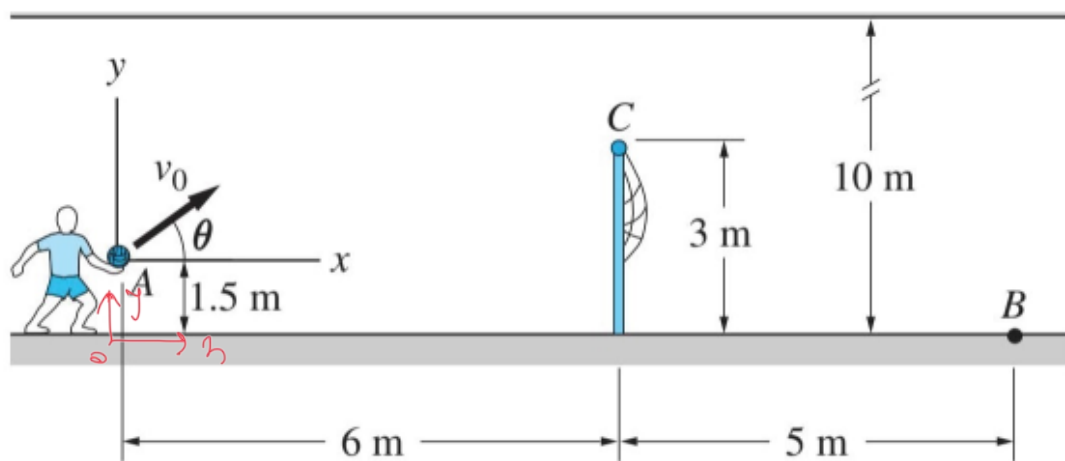




The volleyball player serves the ball from point A with the speed $v_0 = 12 \text{ m/s}$ at the angle $\theta = 28^\circ$ as shown in Figure-1. Neglect the aerodynamic drag for the following calculations.

- (3 marks) What is the velocity of the ball 0.5 seconds after the serve?
- (3 marks) What is the acceleration of the ball at $t = 0.6$ seconds after the serve?
- (4 marks) Determine whether the ball clears the top of the net C and lands inside the baseline B.
- (4 marks) Draw the v_x vs t , v_y vs t , a_x vs t and a_y vs t graphs for the motion of the ball.

Soln: (a) Let's consider the origin at the ground. The velocity of the ball would be

$$\vec{v} = v_x \hat{i} + v_y \hat{j}$$

Now, $v_x = v_0 \cos \theta = 12 \times \cos 28^\circ = 10.59 \text{ m/s}$

$$v_y = v_0 \sin \theta - g t = 12 \times \sin 28^\circ - 9.8 \times 0.5 = 0.73 \text{ m/s}$$

So, $\vec{v} = 10.59 \hat{i} + 0.73 \hat{j}$ Ans

or $|\vec{v}| = 10.61 \text{ m/s}$ $\theta = 3.94^\circ$ Ans

(b) The acceleration of the ball at $t = 0.6$ later is $\vec{a} = -9.8 \hat{j} \text{ m/s}^2$ Ans

Since there is no acceleration along the horizontal direction.

c) to cross the net at certain time the vertical displacement has to be greater than 3m.

For time. $x = v_{0x}t$

$$\Rightarrow t = \frac{x}{v_{0x}} = \frac{6}{12 \cos 28^\circ} = 0.56 \text{ sec}$$

$$\Rightarrow \boxed{t = 0.56 \text{ sec}}$$

let find the vertical displacement at $t = 0.56 \text{ sec}$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$\Rightarrow y = 1.5 + 12 \sin 28^\circ \times 0.56 - 4.9 \times (0.56)^2$$

$$\Rightarrow \boxed{y = 3.11 \text{ m}} > 3 \text{ m}, \text{ so, the ball cross the net}$$

Now, again. $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$

$$\Rightarrow 0 = 1.5 + 12 \sin 28^\circ \times t - 4.9 t^2$$

$$\Rightarrow 4.9 t^2 - 5.6 t - 1.5 = 0$$

$$\Rightarrow \boxed{t = 1.36 \text{ sec}, -0.22 \text{ sec}}$$

$$\text{So, } \boxed{x = v_{0x}t = 12 \cos 28^\circ \times 1.36 = 14.40 \text{ m}}$$

So, the ball clear the net and land outside the baseline (11m).

